The chewing gum myth

CHEAT SHEET

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What is gum made of?



- Sugar, flavors and colors dissolve in your saliva
- The gum base stays: it is soft, rubbery and stretchy
- Old gum came from tree sap (chicle, Maya); today mostly polymers

Polymers



- Polymer = giant molecule made of thousands of small linked pieces
- Long chains tangle like spaghetti, so gum stretches
- Oldest chewed gum: birch pitch from Denmark, about 5,700 years old

How digestion works



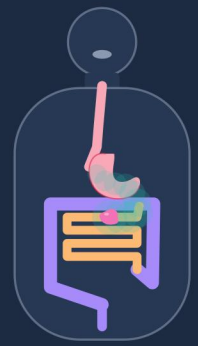
- Mouth › esophagus › stomach › small intestine › large intestine
- Stomach acid is about as sour as lemon juice; enzymes cut food
- The whole journey usually takes about 1 to 3 days

Why gum base is not digested



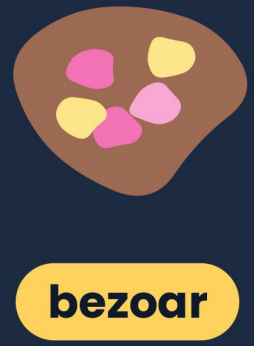
- Enzyme = tiny scissors made of protein
- Each enzyme fits only certain bonds, like a key and a lock
- No enzyme fits the gum base chains, acid can't dissolve them

Not stuck for 7 years



- Your stomach squeezes about 3 times per minute
- Between meals a cleaning wave (migrating motor complex) runs every 1.5 to 2 hours
- It pushes undigested things along: gum leaves in a few days

When it is a problem



- Lots of gum every day can form a hard lump: a bezoar
- Pediatrics, 1998: children who swallowed gum daily needed treatment
- Rule: one piece now and then is OK, young kids no gum

BONUS Undigested does not mean stuck: corn skins and plant fibers also travel through and come out again. Quiz: which part can your body not digest? B: the gum base.

Sources: Milov et al., Pediatrics (1998); Jensen et al., Nature Communications (2019); Mayo Clinic; Harvard Health Publishing

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